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Development of a quartz crystal sensor system to monitor local soil removal during cleaning in closed food processing lines

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ABSTRACT

Closed production lines are often cleaned by oversized cleaning in place (CIP) procedures designed according to worst-case scenarios. This study presents an inline sensor system for monitoring the local cleaning process of swellable food soils based on the inverted piezo electrical effect. The sensor uses quartz crystals to detect soil layers on its surface. A network analyser induces an AC voltage into the quartz and in return measures the resulting resonance parameters. The resonance of the quartz differs with its mass. Hence, any adhering soil layer influences the measured signal.

First tests with the sensor were performed in a flow channel test rig. Therefore, the quartz was coated with a model soil and afterwards cleaned in the channel. As a reference, an established camera sensor monitored the cleaning process. Starch was used as model soil. The tests showed that the sensor is able to detect the soil removal from its surface and that the signal correlates with reference data from the camera sensor. Furthermore, it was possible to observe an influence of the operating parameters, such as fluid temperature or cleaning agent, on the sensor signal, what needs to be compensated by calibration and smart data processing in further studies.

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1. Introduction

To meet hygienic requirements, machines and production lines in the food industry must be cleaned regularly (EU, 2006). Therefore, automated cleaning in place (CIP) systems shall guarantee safe and reproducible removal of physical, chemical and biological hazards. At the same time, the aim is to minimize cleaning time and the use of cleaning chemicals and water (Bremer et al., 2006). But in practice, most systems are designed for the worst-case scenario and follow fixed and oversized cleaning protocols due to a lack of knowledge regarding quantitative prediction of cleaning processes. This excessive effort leads to unnecessarily high production costs (Fryer and Asteriadou, 2009; van Asselt et al., 2002).

Several techniques, such as optical methods (Helbig et al., 2015; Joppa et al., 2019), ultrasound (Withers, 1996), vibration sensors (Da Silva, 2008; Withers, 1996), heat transfer coefficient (Ingimundardottir and Lalot, 2009) or pressure drop monitoring (Riverol and Napolitano, 2005), have already been used to monitor fouling and cleaning processes. The estimation of the fouling state can be either globally or locally. A good overview regarding advantages and disadvantages of such monitoring methods used in heat exchangers is given by Wallhäußer et al. (2012).

Optical methods based on fluorescence e.g. can measure very thin soil layers until few microns (Helbig et al., 2015; Joppa et al., 2019). On the other hand, they are hard to integrate into closed industrial systems in a confined space, since they require a camera, light sources and an optical access and also they are blind for non-fluorescent soil layers. Systems, which measure electrical resistance (Chen et al., 2004) in comparison, can be integrated appropriately in-situ but generate heat while used and may influence the product. Sensor approaches, which measure turbidity or conductivity of the cleaning fluid on the outlet, allow only an indirect conclusions regarding the general cleaning status in the whole system. Local residues in weak spot areas might not be recognized. Conductivity sensors are not suitable for all kinds of cleaning chemicals. The determination of the microbiological status of a surface is another option to validate the cleaning success. Since these tests are time consuming, they are not suitable for a consistent inline use (Winter, 2009). Consequently, inline-capable sensors are suggested, which fulfil the requirements regarding integration and sensitivity of the specific application and which do not influence the product.

The aim of this work is, to present a sensor system based on quartz crystals in order to monitor the removal of swellable soils in closed food productions systems such as pipes. The analysis of thin layers by means of piezoelectric quartz crystals, e.g. quartz crystal microbalances (QCM), is an established laboratory method on various application fields in chemistry, biology or physics (Brünig, 2011; Lucklum, 1997, 2002). Since the first description of the principle (Sauerbrey, 1959), the relevant international state of the art comprises several thousand publications on technical solutions for the control and processing of measured values as well as on the construction of measuring cells for special applications (Lucklum and Hauptmann, 2006; Sauerbrey, 1959). A significant industrial use of QCM apart from research takes place on the field of controlling deposition processes of thin, rigid layers (sputtering, vapour phase deposition, etc.) (Gotthold, 2015). Process measurement technology based on QCM for the inline monitoring of fluids has also been implemented for the measurement of rheological properties of crude oil and lubricants to monitor fuel dilution (Vectron International Inc., 2007). In another application, (Olesen et al., 2016) use a QCM sensor to characterize the cleaning efficiency of different cleaning detergents for a very thin, greasy and marginally swellable soil layer within a research test setup. However, the inline use for monitoring CIP cleaning in food processing plants, as presented in this work, is a new application of quartz crystals. A similar approach is the measurement of cavitation intensity in ultrasonic baths (Jüschke et al., 2014; Koch et al., 2007). In this application the sensor measures the removal of rigid particles from the surface while it is submerged in the cleaning fluid. The same principle is used to measure the slow deposition of aluminium trihydroxide scale particles in heat exchanger applications with a QCM (Brisach et al., 2009). All these cleaning applications with QCM sensors have in common that they use either rigid or very thin and non-swellable viscous soils, which allow a good measurement of the soil layer by frequency analysis. The difference of the application presented in this work is the focus on Type 3 soil layers, according to (Fryer and Asteriadou, 2009), which describe the removal of cohesive solid deposits that require cleaning chemicals to remove them efficiently. In comparison, Type 1 and 2 soil layers describe highly viscous fluids removed with water and the cleaning and killing of biofilms. The swellable properties of Type 3 soil layers will most likely change the sensor signal significantly, since the layer thickness is much higher and viscosity changes significantly during cleaning.

QCM work on the principle of the inverse piezoelectric effect. When a high-frequency AC voltage is applied to a disc of piezoelectric material with a suitable crystallographic orientation (transverse or longitudinal) and gold electrodes applied to both sides, the disc is excited into shear vibration. By measuring the oscillation spectrum in the resonance range, resonance frequency and oscillation amplitude can be determined, among other things. If there is a layer on the disc, it becomes part of the oscillation system. With very rigid layers, especially the resonance frequency changes depending on the layer thickness. With more viscous layers, on the other hand, it is mainly the damping properties and thus the oscillation amplitude that changes. In both cases, conclusions about the layer thickness can be drawn by determining these target values (Brünig et al., 2008; Lucklum, 1997; Windisch and Junghans, 2013). The acoustic transducer principle results in a very high sensitivity of mass coverage (up to 10^{-10} g/cm^2). With regard to cleaning processes, the hypothesis is made that quartz crystal sensors can also be used to detect contamination layers in closed pipe systems in order to monitor the hygiene status or the cleaning progress.

This assumption is supported by research findings in the field of hydrogel sensors, which represent a related application. In this setup the quartz crystal measures the swelling of a submerged viscous hydrogel layer in order to draw conclusions on the cleaning agent concentration (Windisch et al., 2016; Windisch and Junghans, 2013).

To monitor the cleaning process in pipe systems, the sensor has to be implemented in certain local strategic positions, where the cleaning effect is expected to be the worst. It can then be assumed that when those areas are detected as clean by the sensor that the whole system is most likely clean. The sensor surface needs direct contact with the soil and has therefore also be designed according to hygienic design criteria. However, with regard to the concept to measure the cleaning progress in weak spot areas, it is acceptable or in some cases even desirable, if the local cleaning efficiency is slightly decreased by the sensor itself, since the overall cleaning efficiency can be increased significantly due to the additional information from the sensor.

2. Materials and methods

2.1. *General test setup*

For the tests, a sensor prototype (Fig. 1(a)) was developed in order to meet the requirements in the pipe system of a food production line, where high pressures and temperatures can occur. The core of the sensor is a 10 MHz quartz crystal with an AT cut (specification of cut angle through the quartz to realise

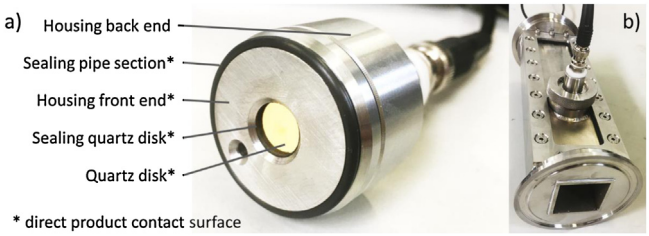


Fig. 1 – (a) Prototype of the used sensor (b) sensor integrated in flow channel section.

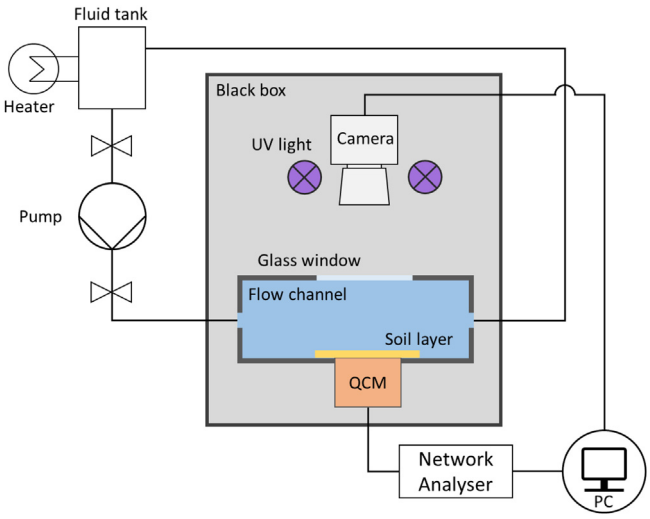


Fig. 2 – Schematic figure of the test setup with the two sensors integrated in a flow channel test rig.

shear oscillation). All used materials are food-safe and suitable for the conditions (high temperatures and pressure, cleaning chemicals), which occur during production and cleaning in the food industry. The design of the casing allows the quartz to perform the shear oscillation appropriately. The sensor was integrated in a rectangular channel section (41 mm × 41 mm, Fig. 1(b)).

The rectangular shape is required by the used CIP test rig. Furthermore, the perspective concept for the sensor is to integrate it in critical areas of a production system, such as flanges or dead ends. Therefore, it is not necessary to integrate it into normal round pipe walls and the presented use for flat surfaces is expedient for most applications.

On the opposite side of the channel is a glass window so that the sensor surface can also be evaluated from outside with a camera sensor. A network analyser (Advantest R3765CG) measures the oscillation parameters of the sensor such as resonance frequency and impedance. The whole test setup was integrated into an existing flow channel test rig (Fig. 2), where the channel section can be mounted and flow parameters such as volume flow rate, temperature and cleaning agent can be adjusted. With the test rig's camera the quartz crystal sensor surface can be monitored inline.

For the measurements with the sensor, the network analyser scans a pre-set frequency range and measures the impedance values. Fig. 3 shows exemplary a resulting curve from this process. By determining the resonance peak it is possible to find the resonance frequency F_R and the associated impedance value Z_R as resulting scalar target parameters in order to compare measurements with different operating parameters.

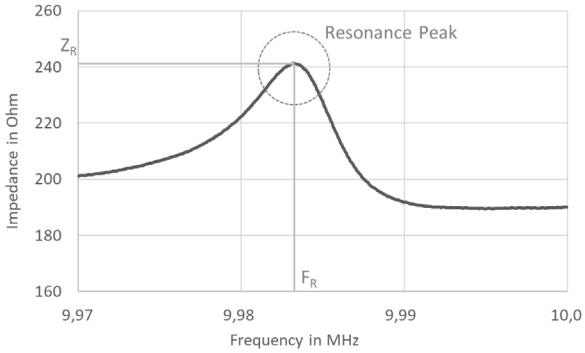


Fig. 3 – Exemplary measurement showing the target parameters Z_R and F_R.

Based on this, the first step was to perform measurements with a clean sensor surface (optically clean und rinsed with ethanol) in order to find the influence of the operating parameters on the sensor signal within the given test setup. Therefore, the following parameters were varied within a full factorial experimental design:

- Fluid temperature: 25 °C, 40 °C and 55 °C
- Mean bulk velocity: 0.4 m/s, 0.5 m/s and 0.6 m/s
- Cleaning agent: deionized water and sodium hydroxide (NaOH, 1 %, v/v)

The fluid temperature was controlled by an instantaneous water heater and constantly measured shortly before the flow channel section. For all operating parameters, the two target parameters resonance frequency F_R and associated impedance value Z_R were compared to quantify the influence. Mean values were determined out of 10 measurements for each parameter setup.

Subsequently, cleaning tests with the sensor were performed to determine its capability to monitor the cleaning progress.

2.2. *Soiling method and cleaning tests*

For cleaning tests, a defined, reproducible and even soil layer was produced on the sensor surface with starch (Cargill C*Gel-Instant, type: 12,410, 150 g starch mixed with 1 L deionized water) as test model soil. The uneven surface of the sensor (Fig. 1(a)) impedes the uniform application of the fluid soil on the sensor by scraping. Therefore, the starch was applied on a flat plastic plate at first and dried there. Due to the surface tension of the plastic plate, it was possible to exfoliate the starch layer (mean layer thickness: 50 µm ± 25 µm) from the plate after drying and to apply it to the sensor by wetting it again (Fig. 4). Subsequently, the cleaning test was performed immediately without further drying. On this way, the soil layer thickness was similar on the whole sensor surface. To make the soil visible for the camera under UV light during the cleaning process, fluorescent stabilised strontium aluminate crystals were added as optical tracer to the starch.

After the soil application, the sensor was mounted in the flow channel test rig in order to start the cleaning test. During the tests, the sensor signal was captured by the network analyser every 10 s. In addition, the cleaning progress on the quartz crystal surface was monitored with the camera sensor to validate the sensor signal (e.g. Helbig et al., 2015; Joppa et al., 2019). During the cleaning test, the cleaning fluid flows constantly over the soiled surface in the channel area and thereby

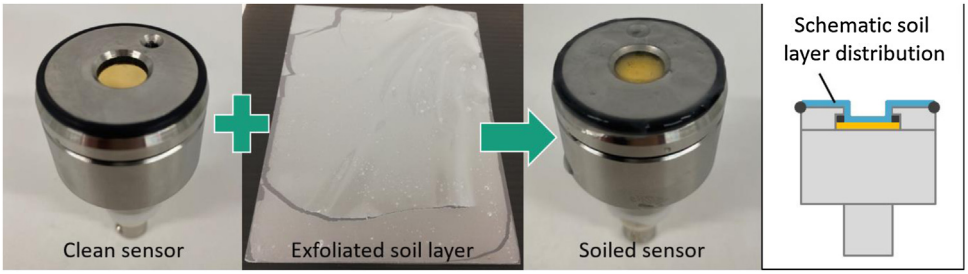


Fig. 4 – Soil application to the sensor surface.

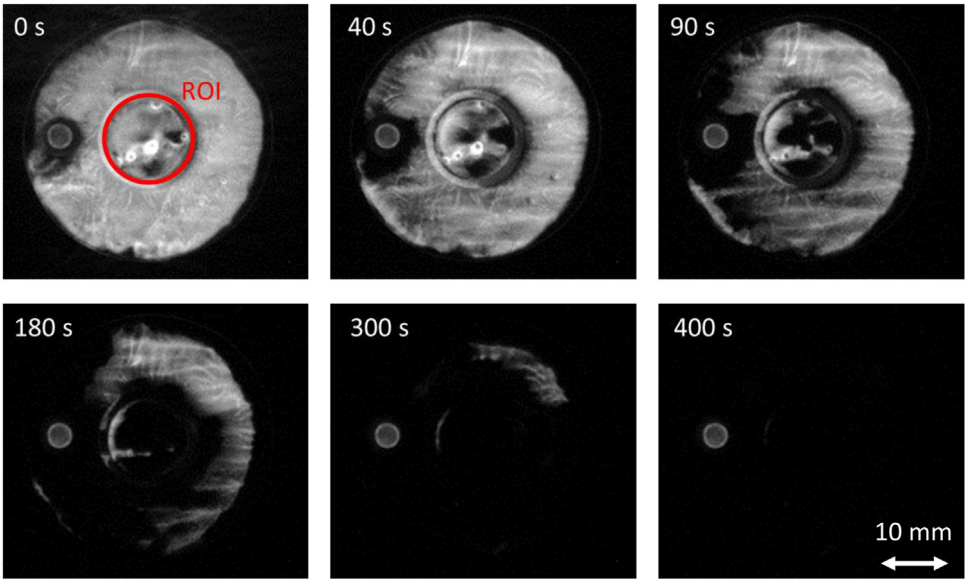


Fig. 5 – Exemplary images taken by the camera sensor at different moments during the cleaning process.

dissolves the soil gradually and removes it due to the shear stress on the surface. The test was stopped after no further soil could be detected with the camera on the quartz surface. The tests were performed with the same set of parameter combinations as the tests with the clean sensor. However, the focus here was not on comparing the cleaning effect for different parameters but on testing if the QCM sensor signal fits with the measurements from the camera.

2.3. *Data evaluation*

To draw conclusions about the residual soil on the sensor surface, the determined resonance values from the operating parameter studies with the clean sensor surface are now used as reference. The values for the clean sensor are subtracted from the measured values during the cleaning test. The resulting difference values are then normalized between 0 and 100% so that they can work as a scale for the residual soil, with 100% referring to a fully soiled surface. To verify the reliability of the data from the quartz crystal, the results are also compared with the measurements from the established camera sensor. Fig. 5 shows exemplary a selection of images that were taken during the cleaning process. Due to the fluorescence the soil appears bright in comparison to the dark surface of the sensor. The local fluorescence intensity, and therefore also the local brightness in the image, correlates with the soil layer thickness. Therefore, the mean grey value in the region of interest (ROI) can be used as a parameter for the relative amount of residual soil. Since the soil layer is already wetted at the beginning of the cleaning test due to the application method, there

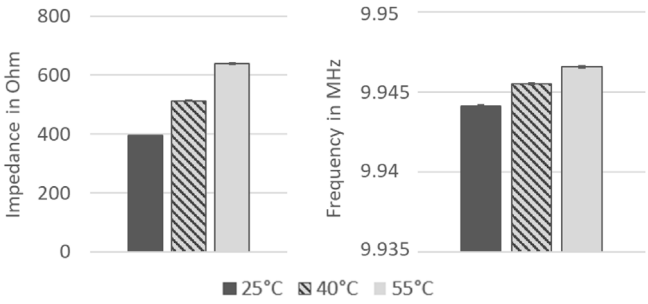


Fig. 6 – Influence of the Fluid temperature on the measured resonance parameters with NaOH (1%, v/v).

is also no initial grey value increase as it was observed in other studies with this test model soil (Joppa et al., 2019). The grey values are extracted from the taken images after conducting the tests by using MATLAB. The used script finds the ROI in every image and calculates the mean grey value in an 8-bit range from all pixels in this area. These mean values are then also normalized between 0 and 100% as it is also done for the data from the QCM sensor. On this way, the results from both sensors can be compared with each other.

3. Results and discussion

3.1. *Parameter studies with clean sensor*

In the first step, tests with the clean quartz sensor were conducted according to the experimental design. Fig. 6 shows the results of the tests in NaOH for different temperatures.

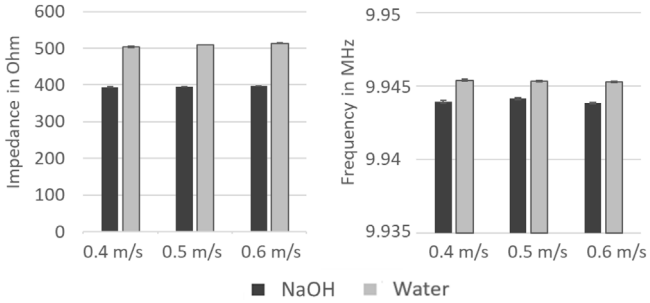


Fig. 7 – Influence of the cleaning agent and the flow rate on the measured resonance parameters (temperature: 25 °C).

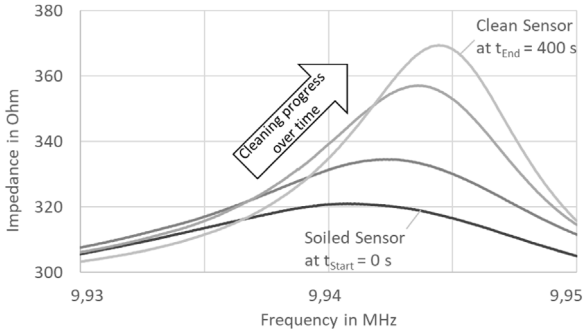


Fig. 8 – Shift of the measured sensor curves over cleaning time (exemplary for one cleaning test with 1% [v/v] NaOH at 25 °C).

Both, resonance frequency and impedance, increase together with the temperature. It is assumed that this effect is caused by the general influence of temperature on the quartz crystal behaviour as well as by the reduced fluid's viscosity due to the higher temperature (Lucklum, 2002; Sauerbrey, 1959; Zeitvogel, 2009). In water, the measurements showed the same linear tendency.

Fig. 7 shows the influence of cleaning agent and flow rate on the measured resonance parameters. For the flow rate, there was no significant change in the signal which could be observed. The cleaning agent in contrast leads to different measured values for both target parameters. As also for the temperature, it is assumed that this is a result of the higher viscosity of NaOH solution in comparison to water (Lucklum, 2002; Zeitvogel, 2009). Therefore, the damping of the oscillation is less when water is used.

In an industrial production and cleaning process, all kinds of cleaning agents can be used. Therefore, a suitable calibration is required, which is performed under real operating parameters to identify the correct resonance parameters for a clean sensor surface so that these values can be used as a reference.

3.2. *Cleaning tests*

In the second step, cleaning tests with a soiled sensor surface were performed for the different operating parameter setups. Fig. 8 shows exemplary a clear change of the resonance values during the cleaning process in NaOH at 25 °C. The soil layer on the quartz crystal attenuates the oscillation what leads to a decrease of the impedance. But also the resonance frequency slightly shifts due to the decreased mass.

For all measured curves during the cleaning process, the target parameters F_R and Z_R were determined, referenced to the signal for a clean sensor, normalized between 0 and 100 % and compared with the signal from the camera sensor. Fig. 9

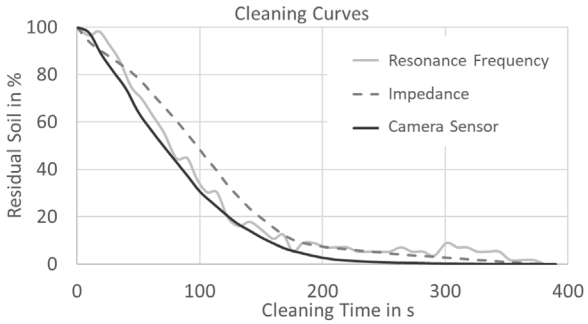


Fig. 9 – Shift of the measured residual soil depending on the target parameter comparing the quartz crystal sensor with the camera signal.

shows the resulting cleaning curves in comparison. There is a good correlation between the signal from the camera and the data from the quartz crystal sensor. The impedance seems to be the more suitable value since the resonance frequency is more unstable and noisy. This is probably a result of the relatively flat curves measured with a soiled quartz sensor. Small deviations while determining the curve maximum can therefore result in a relatively high frequency shift, while the impedance is barely affected due to the curve shape. The primary aim of the sensor is to determine the end of the cleaning process. Therefore, a less noisy signal is desirable.

4. Conclusions

The results suggest that with the presented quartz crystal sensor it is possible to monitor the cleaning progress of a surface soiled with starch. Thereby, the impedance seems to be the more suitable target value because it is less noisy than the resonance frequency.

Since temperature and viscosity of the cleaning fluid have a significant influence on the oscillation of the quartz crystal, it is important to calibrate the system according to the operating parameters used during the cleaning progress. In an industrial environment the sensor could for example be calibrated by measuring the signal with a clean surface at the end of a regular cleaning process at different temperatures in order to set itself for the specific operating parameters and fluid viscosity.

Further tests with different test soils should be aimed for in order check if the quartz crystal sensor also works for substances with a different cleaning and swelling behaviour.

In addition, for cases where the operating parameters and cleaning fluids are varying or unknown to the sensor, it is recommended to do further examinations regarding alternative target parameters and analysis methods of the measured data in order to differentiate between a soiled and a clean sensor surface. Possible alternative target parameters are for example the width of the impedance curve at a certain height or the intersection of the turning tangent with the frequency axis (Jüschke et al., 2014).

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Conflict of interest

The authors declare that they have no conflicts of interest.

Declaration of Competing Interest

The authors report no declarations of interest.

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